EARTH AND MINERAL SCIENCES

Lumpy, bumpy hail: Realistic hail shapes may improve modeling of severe weather

A gargantuan hailstone that fell in Argentina may have set a world record, according to researchers. **Credit: Victoria Druetta . All Rights Reserved.**

EXPAND +

OCTOBER 22, 2024By Matthew Carroll

UNIVERSITY PARK, Pa. — Though often compared in size to round objects — peas, golf balls or even softballs — hailstones do not fall from the sky as smooth spheres. A new approach for modeling hailstorms that uses more realistic hailstone shapes could improve our understanding of hazardous weather, according to a team led by scientists at Penn State.

They reported their findings in the **Journal of the Atmospheric Sciences**.

“In numerical modeling of hailstorms, the scientific community usually considers hailstones to be spherical to simplify calculations,” said Yuzhu Lin, a doctoral candidate in the Department of Meteorology and Atmospheric Science at Penn State who led the study. “But in nature, hailstones generally are not spheres and can be rather lumpy and have spikes.”

Swapping out smooth spheres for naturally shaped hail impacted how the hailstones moved through storms in the models, resulting in changes in the size of hail and the location where it fell, the scientists said.

“If you model an idealized spherical hailstone, the growth will be rather unrealistic and it may lead to you conclude that a certain kind of storm environment will grow big hailstones, and that might not actually be true,” Lin said. “Our work shows that changing hail characteristics in these models changes their movement and how they interact with the environment as a result.”

Instead of dropping like a bowling ball, hailstones tumble about through storm clouds as they fall, the scientists said. Their shape impacts factors like heat transfer between the hail and the environment and the collection of hydrometeors in the air — like water droplets and ice particles. This can impact how the hail grows and the path it takes through the storm.

“Even small changes can alter the trajectory of hailstones at the same starting point,” Lin said. “Once you alter a small thing then they’re growing differently, and they will move differently in the storm. It’s like the butterfly effect, it changes everything.”

The scientists tapped into a database created by researchers with the University of Queensland in Australia, who have collected and logged data from hailstones across Australia.

The researchers used the hail database to populate their model with random realistic hailstones. They included random tumbling of individual hailstones to simulate free-falling behavior.

“For our models to better understand what happens in the real world, we need to have good data on the size and mass of realistic hailstones,” Lin said. “And if we can improve our models, we’ll have more skillful warnings as a result.”

Further research is needed to investigate the true tumbling behavior of hailstones during storms, the scientists said. This additional data could lead to further improvements in their numerical model’s ability to handle highly complex hailstone physics.

“We hope to better identify the storm environments that can potentially produce severe hail and, or large amounts of hail,” said Matthew Kumjian, professor of meteorology at Penn State, Lin’s adviser and a co-author of the study. “If we can improve hail forecasts, we can help the public stay safe and mitigate damage during hailstorms and aid hail-sensitive industries like insurance and agriculture.”

Joshua Soderholm, an honorary senior research fellow at the University of Queensland, and Ian Giammanco, lead research meteorologist with the Insurance Institute for Business and Home Safety, also contributed.

The U.S. National Science Foundation and the Insurance Institute for Business and Home Safety supported researchers on this project.

LAST UPDATED OCTOBER 22, 2024

CONTACT

Patricia Craig

plc103@psu.eduCell Phone: 814-863-4663http://www.ems.psu.edu

TAGS

- Research
- Earth and Environment
- Students
- Visitors and Neighbors
- Faculty and Staff
- News of Record
- Publications
- Business and Industry
- University Park
- Earth and Mineral Sciences
- Department of Meteorology and Atmospheric Science
- Fox Graduate School
- Latest News

Phone: 1-814-865-4700

The Pennsylvania State University
201 Old Main
University Park, PA 16802

Explore

News

Careers

Colleges

Campuses

Contact Us

Student Support

Registrar

Bursar

Student Affairs

Housing and Food Services

Quick Links

Maps

Directory

Libraries

Academic Calendar

LionPATH

Penn State Health

Resources

Prospective Students

Current Students

International Students

Business and Industry

Veterans and Military

Visitors

Faculty and Staff

Alumni

Media

Stay Connected

Emails / Headlines

Emergency Notifications

Penn State Go

Strategic Communications

Report Misconduct

Police

The Pennsylvania State University Privacy Statements Non Discrimination Accessibility Legal Statements

The Pennsylvania State University © 2025